

AIS Glass — Data Validation and Forecast Report

Demand Forecast & Inventory Intelligence — Proof of Concept

Prepared Oct 6, 2026

Purpose

This report answers one question: do the numbers shown in our proof of concept agree with the data AIS Consumer Glass Solutions supplied?

They do. Every unit figure reconciles exactly across all 53 branches and all 2,063 ordered SKUs. Order value reconciles exactly at 51 of the 53 branches; the remaining difference is ₹1,66,548, or 0.019% of the network's order value, and it is traced to two specific order lines.

The reconciliation did not pass on the first attempt. Three genuine defects in our processing were found and corrected during this exercise; they are documented in full rather than omitted. Two further apparent mismatches turned out to be errors in the verification itself, and those are documented too.

The report also sets out what the client data contains, what we hold in the system, and what we forecast, so that the reconciliation can be read against a clear picture of the inputs.

The data AIS supplied

Five files, delivered as one archive. Every file is held read-only; nothing in the system writes back to any of them.

File	Format	Sheet	Rows	What it holds
Orders & Receipts (Lead Time)	.xlsx, 64 MB	Receipts (Lead Time)	775,912	One row per order line. The demand record.
Sales Data FY 24~26	.xlsb, 79 MB	FY 25-26 Sales	1,703,042 across both	One row per invoice line, current year
		FY 24-25 Sales		Same, prior year

File	Format	Sheet	Rows	What it holds
Stock in Hand as on 1st Aug'26	.xlsx, 6.7 MB	Closing Stock Jul'26	144,921	Stock position at a single date
Substitution Mapping	.xlsx, 218 KB	Sheet1	2,417	Product master and substitute parts
Location Master	.csv, 24 KB	—	57	Branch master, lead times, service factors

What each sheet actually contains

Receipts (Lead Time) — the order book, and our forecasting target. Eighteen columns. The ones that matter: **Depo** (ordering branch), **Oracle No** (the SKU, 2,063 distinct), **Order Date**, **Despatch Date**, **Quantity** (ordered), **Despatch Qty** (served), **MRP Rate** and **MRP Value**. This single sheet carries both sides of the fill-rate question — what a branch asked for and what it received — which is why it is the primary source rather than the sales file.

FY 25-26 Sales and FY 24-25 Sales — invoiced history. Thirty-three and thirty-four columns respectively; the prior-year sheet carries one extra column, **Doc Series**. These record what was billed, not what was requested, so they cannot measure shortfall. We use them only to extend history backwards into the period before the order book opens. Both sheets carry customer names, GST numbers and consignee details; all of it is dropped at ingestion and never reaches the model, the database or any screen.

Closing Stock Jul'26 — one snapshot, not a history. Branch, product, closing quantity, closing value, MRP value and last GRN date, as at 1 August 2026. Row 1 of this sheet is a title row carrying **SUBTOTAL** formulas rather than data, so the real header is on row 2. Because this is a single snapshot and not a stock ledger over time, inventory recommendations are current-position estimates. We cannot and do not claim a historical backtest of inventory policy — there is no stock history to run one against.

Sheet1 of Substitution Mapping — the product master. Item code, product name, up to two substitute codes, AIS and AFM price points effective 11 May 2026, and the attributes every product chart is built from: **Glass Type**, **Product Group**, **Vehicle Category**, **Vehicle Age Category**, **OEM/Non-OEM** and sales category A/B/C/D.

Location Master — branch reference data. Forty-three columns including region, zone, supply hub, tier, **Transit Lead Time**, **Avg Lead Time**, **Std. LeadTime**, **Service Factor** and **Truck (MoQ)**. Row 1 is a **Branch Master** title row, so the header is on row 2. The file also carries PAN, GSTIN, CIN, TAN, contact names, email addresses, phone numbers and postal addresses, all of which are dropped at ingestion.

Two structural points worth recording

The **Replenishment A**, **Replenishment B** and **Replenishment C** columns in Location Master are **zero for all 57 branch records**. They are therefore never used as a minimum order quantity; doing so would silently set every reorder to nil.

The raw **Avg Lead Time** field contains **0000-00-00** entries and values that produce negative lead times. Lead time is computed on cleaned despatch dates only, and the exclusions are reported rather than absorbed.

What we hold, and what we forecast

The target is ordered quantity, not invoiced quantity. This is a deliberate choice. An invoice records what was shipped, which is already constrained by what was in stock — forecasting it teaches the model to predict our own supply failures. The order line records what the branch actually asked for. Where the order book does not reach back far enough, invoiced sales are used to extend history, and every such row is labelled a sales proxy. The two are never added together or treated as the same measurement.

The grain is one branch × one SKU × one ISO week. A week is reported under the month containing its Thursday, which is the ISO standard rule. No week is split across two months, so weekly totals add back to monthly totals exactly. Daily is not offered, because splitting a week's units across its days would invent values the source does not contain.

Scope

	Covered	Of
Demand analysis — branches	53	53
Demand analysis — SKUs	2,063	2,063
Forecasting — branches	2 (Bengaluru, Delhi-1)	53
Forecasting — SKUs	136	2,063
Forecast lines (branch × SKU)	261	

The demand analysis covers the client's entire dataset. Forecasting runs on a deliberately narrowed pilot of two branches and 136 SKUs — enough to prove the method at a cost that fits a proof of concept. Every screen states its own scope on the page; none of them implies network coverage it does not have.

The models

Thirteen model families compete for every single line, independently. There is no one chosen algorithm applied across the board.

SARIMAX · SARIMAX with exogenous variables · Auto ARIMA · Auto ARIMA with exogenous variables · XGBoost · XGBoost with exogenous variables · Exponential Smoothing in four variants (additive, additive damped, multiplicative, multiplicative damped) · VAR · VAR with exogenous variables · LSTM.

The most recent run fitted **4,779 model runs across 281 series in 19 minutes 39 seconds**. 3,754 completed and 1,025 were ineligible — a model whose data requirements a short series cannot meet. **Nothing failed and nothing timed out**. An ineligible model stays visible on the leaderboard with the requirement it missed; it is never hidden and never silently replaced by a zero forecast.

Four simple benchmarks — naïve, seasonal naïve, and 3- and 6-week moving averages — run alongside as a sanity check. They are never allowed to win, but where one beats the chosen model that is reported rather than suppressed.

What comes out

For each of the 261 lines: a point forecast per week, aggregated to months; demand quantiles at the 80th, 90th and 95th percentiles; and a recommended order quantity built as `order-up-to level - usable stock - confirmed on-order + backorders`, where the order-up-to level is the 95th-percentile demand over a protection period of review period plus branch lead time.

The forecasts across branches, SKUs and the national total are reconciled so that the parts add up to the whole. The current run reports full coherence, with maximum incoherence of 0.0.

How the validation was done

The method matters here, because the obvious way of checking proves nothing.

Reading our own processed output and comparing it against our own screens would only confirm that our code is internally consistent. If a rule is wrong, both sides are wrong in the same way and the check passes. So every comparison figure was **recomputed from scratch, directly from the client's workbooks**, by a separate script that shares no code with the application — it reads the raw sheets, applies the arithmetic independently, and knows nothing about how the system stores or derives anything.

Confirming it is the same data

Before comparing anything, the five files held by the system were checked against the client's original archive by CRC-32 checksum and byte length. All five are byte-for-byte identical:

File	Bytes	Match
Location Master.csv	24,477	Yes
Orders & Receipts (Lead Time).xlsx	67,571,606	Yes
Sales Data FY 24~26.xlsb	82,429,868	Yes
Stock in Hand as on 1st Aug'26.xlsx	6,995,927	Yes
Substitution Mapping.xlsx	223,570	Yes

This removes the most mundane explanation for any difference — that we were working from a different or partially updated copy.

The comparison window

All comparisons are over the order book: **April 2025 to July 2026, 70 ISO weeks**, from the earliest to the latest order date in the file. The system holds history back to April 2024 through the invoiced-sales proxy, so comparing an unfiltered system total against the order workbook would overstate our side by design. Every figure below is filtered to the window the order book actually covers.

What was compared

Not only the headline totals. A grand total can agree while the breakdowns beneath it are wrong in offsetting ways, so each chart was reconciled separately: every one of the 53 branches, every glass type, every value class, every vehicle category, the top SKUs by volume, and every month of the demand trend.

Headline reconciliation

The client's entire network — 53 branches, 2,063 SKUs

Measure	AIS source files	Our analysis	Status
Order lines read	775,912	775,912	Exact

Measure	AIS source files	Our analysis	Status
Units ordered	2,602,392	2,602,392	Exact
Units despatched	2,133,961	2,133,961	Exact
Units short	504,052	504,052	Exact
Fill rate	82.00%	82.00%	Exact
Order value	₹8,571,223,655.30	₹8,571,057,107.30	₹1,66,548 apart
Branches	53	53	Exact
SKUs ordered	2,063	2,063	Exact
Order window	1 Apr 2025 – 31 Jul 2026	1 Apr 2025 – 31 Jul 2026	Exact

These are not rounded figures chosen to look tidy. Two million six hundred and two thousand, three hundred and ninety-two units on both sides, from two separate calculations over the same file.

The forecasting pilot — Bengaluru and Delhi-1, 136 SKUs

Measure	AIS source files	Our analysis	Status
Units ordered	146,814	146,814	Exact
Units despatched	129,559	129,559	Exact
Units short	20,449	20,449	Exact
Fill rate	88.25%	88.25%	Exact
Order value	₹59.95 crore	₹59.95 crore	Exact

On the pilot scope every figure reconciles exactly, value included. The two lines responsible for the network value difference both fall outside these two branches.

Value: the ₹1,66,548

That gap is 0.019% of order value and comes from exactly **two lines pulling in opposite directions**. It is set out in full further on. No other branch differs by a single rupee: **51 of the 53 branches reconcile exactly on value**, and all 53 reconcile exactly on every unit measure.

Chart by chart

A grand total can agree while the breakdowns underneath it are wrong in ways that cancel out. Each chart was therefore reconciled on its own.

By branch — all 53

Every branch reconciles exactly on units ordered and units despatched. No exceptions, no tolerances applied. On value, 51 of 53 reconcile exactly; Jabalpur and Coimbatore are covered in the next section.

By glass type

Glass type	AIS source files	Our analysis
Lam	1,674,670	1,674,670
Sidelite	577,327	577,327
Backlite	347,748	347,748
Not in product master	2,641	2,641
Temp W/s	6	6
Total	2,602,392	2,602,392

By value class

Class	AIS source files	Our analysis
A	1,919,460	1,919,460
B	440,196	440,196
C	173,231	173,231
New Model	61,085	61,085
D	5,779	5,779
Not in product master	2,641	2,641
Total	2,602,392	2,602,392

By vehicle category

Category	AIS source files	Our analysis
Car & MUV	2,234,409	2,234,409
Commercial	257,626	257,626
3W	104,998	104,998
High End	2,718	2,718
Not in product master	2,641	2,641
Total	2,602,392	2,602,392

One point worth recording here. The product master spells this category two different ways — Car & MUV on 1,530 SKUs and Car & Muv on 539. Read literally, those are two separate categories and the chart would split one real category into two bars. We treat them as one, which is why the comparison above reconciles. The same applies to Commercial and High End.

By SKU

All 2,063 ordered SKUs are present on both sides, and the top ten by volume reconcile exactly, in the same rank order:

SKU	AIS source files	Our analysis
FG.MP8.LFH.GCG2120000	92,636	92,636
FG.MM7.LFH.GCGB1B0000	64,745	64,745
FG.TCF.LFH.GCG2120000	61,653	61,653
FG.BA5.LFH.GCG2120000	61,614	61,614
FG.MYA.LFH.GCG2120000	41,246	41,246
FG.MPH.LFH.GCG21X0000	39,890	39,890
FG.MY7.LFH.GCG2120000	39,647	39,647
FG.M1C.LFH.GCG2120000	34,862	34,862
FG.MYS.LFH.GCG2120000	34,847	34,847
FG.ME3.LFH.GCG2120000	34,126	34,126

Demand trend

All 16 months of the demand trend reconcile exactly, month by month.

This one needed care. The data is held in ISO weeks, and a week that straddles a month boundary has to be assigned to one month or the other. The rule applied is the ISO standard: the week belongs to the month containing its Thursday. Compared on that basis, all 16 months agree. Compared against plain calendar months, every month disagrees — which is a difference in convention, not in the data, and is addressed in the next section.

The two lines behind the value difference

The whole ₹1,66,548 comes from two lines. Both are traced to the individual order. They pull in opposite directions, which is why the net figure is smaller than either.

Jabalpur — one line where the client's own columns disagree

Field	Value
Branch	Jabalpur
Order number	E1/05-25/02/JLR
Order date	1 May 2025
SKU	FG.VL1.LFH.GCG3130000
Quantity	22
MRP Rate	₹21,500
MRP Value as stated in the file	₹7,09,500
Quantity × rate	₹4,73,000
Difference	₹2,36,500

The stated value of ₹7,09,500 is exactly $33 \times ₹21,500$. The quantity column says 22. The two columns in the client's own file contradict each other on this line, and there is no way to tell from the file which is correct.

We compute value as quantity × rate, so our figure for Jabalpur is **₹2,36,500 lower** than the stated column. We have not adjusted this to match, because doing so would mean overriding a quantity with a value we cannot verify.

This is the only line in all 775,912 where the client's stated value and their own quantity × rate disagree.

Coimbatore — 44 lines with no price at all

44 order lines at Coimbatore, covering 246 units, carry no MRP rate. Their stated value in the file is zero.

We value those units at the average price of the other lines for the same product, branch and week. That adds **₹69,952**, so our Coimbatore figure sits **above** the client's.

The reasoning is that a unit which was ordered is real demand, and pricing it at the rate its neighbouring lines carry is more defensible than recording it as worth nothing. The alternative — valuing it at zero — would understate demand value at that branch and would be equally an assumption, just a less visible one.

The same 44 lines also carry no despatch quantity. Both facts point the same way: these look like incomplete records rather than genuine zero-value orders.

Net effect

	Effect on our figure
Jabalpur, 1 line	–₹2,36,500
Coimbatore, 44 lines	+₹69,952
Net	–₹1,66,548

Against a network order value of ₹857 crore, that is 0.019%. Neither adjustment touches any unit figure, and neither falls within the forecasting pilot.

Defects found during validation, and corrected

The reconciliation did not pass on the first attempt. Three real defects surfaced, all of them silent — none produced an error, a warning or an obviously wrong-looking screen. They are recorded here in full.

1. Average price was not weighted by quantity

Where a branch ordered the same SKU more than once in a week at different prices, we were averaging those prices with equal weight. A line for 2 units counted as much as a line for 2,000.

Every value on every screen is computed as ordered quantity $\times$ average price, so wherever a cell held more than one price, that product disagreed with the sum of the underlying lines. **19,359 of 582,324 cells hold more than one price.** The network order value was overstated by **₹2.92 crore.**

The fix is to accumulate price $\times$ quantity and quantity separately, and divide at the end. Value now reproduces the line-level sum exactly.

2. A blank despatch quantity was read as zero

When the despatch column is empty, that means the despatch was never recorded. We were treating it as zero despatched, which turned the entire ordered quantity into shortfall.

44 order lines are affected, contributing **246 units of shortfall that never existed** and depressing the fill rate correspondingly. Blank and zero are now distinguished: a blank is counted as unknown and excluded from the fill-rate ratio rather than being scored as a complete failure to supply.

3. Products absent from the product master were being dropped

Two SKUs that AIS orders **across 44 of its 53 branches** do not appear in the Substitution Mapping product master. They therefore have no glass type, no value class and no vehicle category to group by, and our charts were discarding those rows entirely.

The effect was a chart that did not add up to its own headline figure. Every product-attribute breakdown fell **2,641 units short** of the total shown directly above it, with nothing on screen to explain the gap. Anyone adding up the bars would have found the discrepancy and had no way to account for it.

Those rows now appear under an explicit label, **Not in product master**, so they are visible, counted, and the breakdown reconciles to the total. The label names the actual cause rather than hiding the rows under "Other".

Why these did not recur

Each fix is held in place by an automated test that asserts the property that was broken — for instance, that a product breakdown sums to its own total — rather than simply restating the new code. One of the tests fails deliberately against the old behaviour. The full suite stands at **1,093 backend and 268 frontend tests, all passing.**

Forecast accuracy

This is a different question from whether the data reconciles, and the two should not be conflated. The historical data matches exactly. A forecast is a prediction about months that have not happened, and it carries error.

Measured on 261 branch × SKU lines

Forecast horizon	Accuracy	Error (MAPE)	Lines meeting 85% target
One month at a time	71.45%	28.55%	28 of 261
A 3-month total	82.95%	17.05%	104 of 261
A 6-month total	89.55%	10.45%	159 of 261
All SKUs combined, one month	92.54%	7.46%	—

249 of 261 lines improve as the horizon lengthens, and 10 get worse. This is the expected pattern: week-to-week noise at an individual part number averages out over a planning quarter. It is also the practical point — a replenishment decision is made over a protection period of weeks, not for a single isolated week.

How this was measured, and how it was not

Every figure above comes from **rolling-origin validation on held-out history**. The model is fitted on data up to a cut-off, asked to predict forward, and scored against what actually happened. The origin then rolls forward and the process repeats. All imputation, scaling and parameter tuning is refitted inside each fold.

A random train/test split was never used. On time-series data a random split lets the model see future periods while training, and produces a flattering number that collapses in production.

Historical-only columns are never treated as known future inputs. MRP is used to value past demand; its future value is not known, so it is never supplied to a model as a forward-looking driver.

Stated limitations

40 of 272 lines are beaten by a simple benchmark. On those specific series a moving average or a naïve carry-forward currently does better than the fitted model. This is reported rather than suppressed, and it marks where the modelling earns its keep and where it does not yet.

Accuracy and error are reported separately and neither is derived from the other.

Accuracy is floored at zero, so on a line whose error exceeds 100% the subtraction would print exactly 0% and conceal the real figure. Four lines in this run read 0% six-month accuracy against measured errors of 102.8%, 140.8%, 357.5% and 100.0%. Both numbers are carried on every payload and printed on both screens.

Error deterioration over time has not been measured. The forecast origin is the last observed month, so no post-forecast actuals exist yet to compare against. The relevant screen states this rather than showing an empty chart.

Inventory recommendations are current-snapshot estimates. Stock is a single position dated 1 August 2026 while demand history ends in July 2026. There is no stock ledger over time, so no historical backtest of inventory policy is claimed.

Conclusion

The proof of concept reproduces the client's data. All unit figures — ordered, despatched, shortfall and fill rate — reconcile exactly at network level, at every one of the 53 branches, and across every product breakdown and every month of the demand trend. Order value reconciles exactly at 51 of 53 branches, the remaining ₹1,66,548 (0.019%) arising from two identified order lines.

The forecast reaches 89.55% accuracy over a six-month planning horizon on the 261-line pilot, measured on held-out history with a leakage-safe validation design.

Three points to raise with AIS

All three are gaps in the supplied data that this reconciliation brought to light. None blocks the proof of concept, and all three are worth confirming before the system is scaled.

1. **44 order lines at Coimbatore carry neither a price nor a despatch quantity.** Is there a source for these values, or should the 246 units be treated as incomplete records? We currently value them at the going rate of neighbouring lines, which adds ₹69,952.
2. **One line at Jabalpur states a value its own quantity does not support.** Order E1/05-25/02/JLR records 22 units at ₹21,500 but a value of ₹7,09,500, which is 33 units' worth. Which column is correct?
3. **Two SKUs ordered across 44 branches are absent from the Substitution Mapping product master.** They account for 2,641 units in the comparison window. Should they be added to the master, or are they intentionally held outside it?

What a production rollout would need

- **Stock history rather than a single snapshot.** The current recommendations are sound as a present-position estimate, but a stock ledger over time would allow the inventory policy itself to be backtested instead of only the demand forecast.
- **A goods-receipt date.** The order file records order date and despatch date, so the leg from despatch to receipt is not observable. What is currently measured is one leg of lead time, not the whole of it.
- **A decision on lead-time variability.** Standard deviation of lead time is captured per branch but does not currently feed safety stock. North 24 Parganas, for example, shows a 3-day mean with 2.19 days of spread — 73% variability — and is presently given the same cover as a branch with stable lead times. Feeding this in would change every recommended quantity, so it is a commercial decision rather than a technical correction.
- **Extending beyond the pilot.** Forecasting currently runs on 2 branches and 136 SKUs. The full network of 53 branches and 2,063 SKUs is already ingested and validated; scaling the forecast is a question of compute time, not of method.